



ANAMNESIS — a medical chatbot that asks the right next question

Information-gain driven symptom triage over 41 diseases and 132 symptoms

Project-Based Learning · Machine Learning · Department of Computer Science and Engineering · Chennai Institute of Technology · 2026–2027

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01 Problem

Symptom checkers name the right condition first only about a third of the time [Semigran 2015]. Student projects on the standard Kaggle dataset claim 100% — but patients start a conversation with two symptoms, not a 132-field form.

02 Leakage audit — our first finding

304

unique rows in 4,920

100%

of official test rows already in training

61.8%

LR accuracy from 2 symptoms on unseen profiles

The famous 100% is memorisation. We split by unique profile, so no test patient has a copy in training.

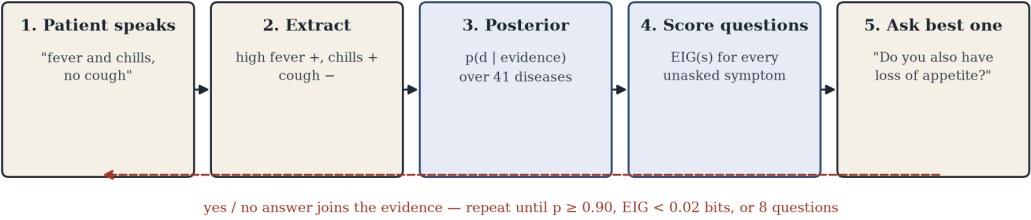
03 Objectives

- Repair the evaluation protocol (grouped split)
- Measure accuracy under partial evidence
- Ask each follow-up question by expected information gain
- Lay-language NLU with negation; warning-sign triage
- No generated text — nothing can be hallucinated

04 Iterations

Version	From 2 symptoms	Top-1
1· Logistic regression	one-shot	61.8%
2· Masked-augmented LR	one-shot	79.7%
Final· Evidence model + EIG	+ 4.0 questions	96.3%

05 Method

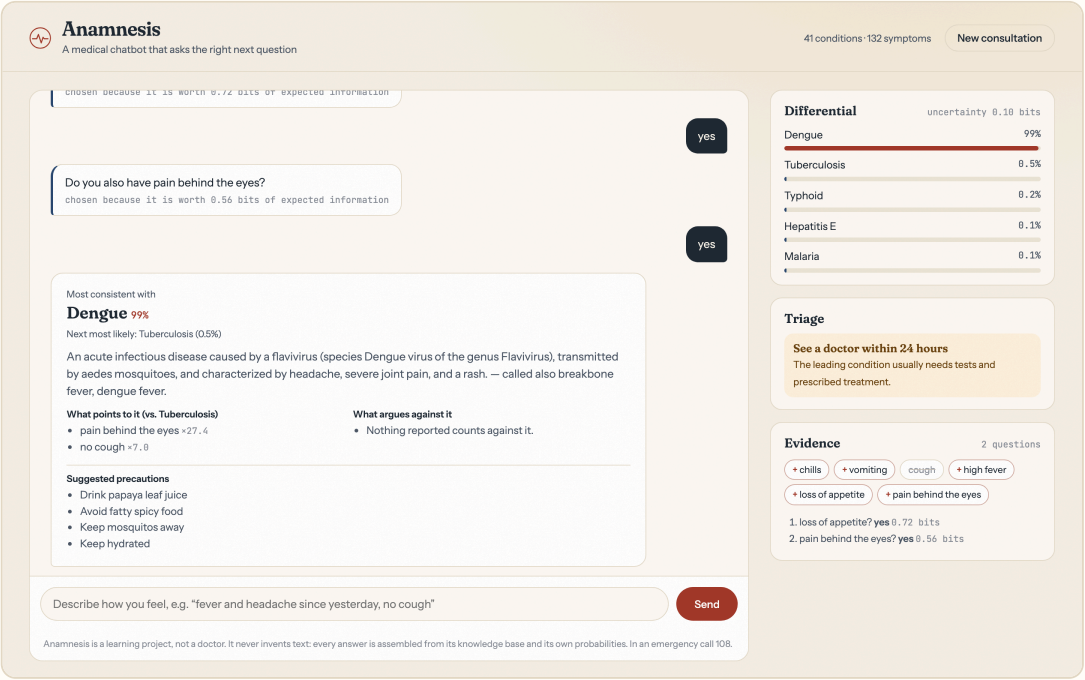


$$EIG(s) = H(D) - [P(\text{yes}) H(D \mid s = \text{yes}) + P(\text{no}) H(D \mid s = \text{no})], \quad P(\text{yes}) = \sum_d p(d) \theta_{ds}$$

$$p(d \mid E) \propto p(d) \cdot \Pi_{\text{present}} q_{ds} \cdot \Pi_{\text{absent}} (1 - q_{ds}) - \text{unknown symptoms drop out}$$

Each turn scores all 132 unasked symptoms by expected entropy reduction (0.18 ms) and asks the best one; stops at 90% confidence, abstains below 50%.

06 The chatbot



07 Results — 1,230 simulated consultations, unseen profiles

96.3%

top-1 accuracy

98.7%

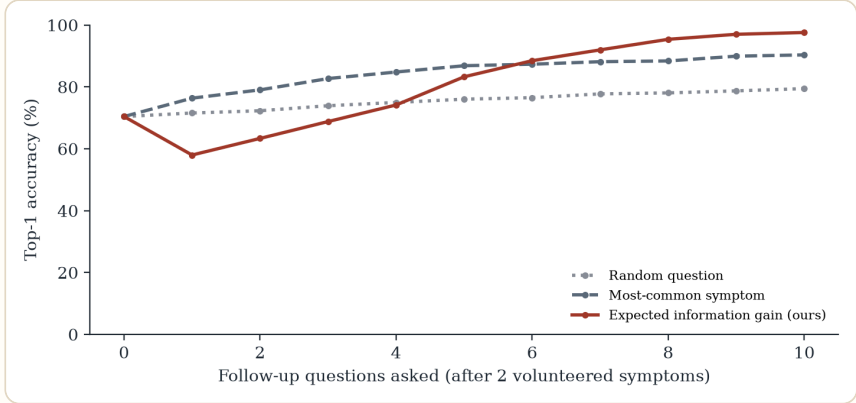
top-3 accuracy

4.0

questions on average

3% noisy answers · ECE 0.035 · NLU F1 0.927 on 50 lay-language utterances

08 Why information gain



Policy	Top-1	Questions
Random question	79.4%	7.9
Most-common symptom	90.4%	5.2
Expected information gain	96.3%	4.0

09 Conclusion & future scope

Well-chosen questions recover the accuracy that partial information takes away. Next: clinician review, seasonal prevalence priors (dengue in monsoon), Tamil input, real-user pilot.